ADITYA KULKARNI

AI/ML & Backend Development

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SUMMARY

AI/ML and Backend Software Engineer with hands-on experience designing REST APIs, AI agents, and full-stack applications using Python, Node.js, Flask, React, and SQL. Skilled in building and deploying machine learning models, integrating LLM-based agents via the Model Context Protocol (MCP), and shipping production-style backend systems with clean, testable architecture. Strong foundation in data structures, algorithms, and Agile development practices, with experience in building end-to-end applications from database schema to UI.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C,C++, SQL

Frameworks & Libraries: React.js, Flask, Node.js, Express.js, REST API Design, HTML5, CSS3, Tailwind CSS

AI/ML: Machine Learning, Generative AI, AI Agents, Model Context Protocol (MCP), Prompt Engineering, Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib

Databases: MongoDB, MySQL, SQLite

Tools & Practices: Git, GitHub, VS Code, Postman, API Integration, Data Structures & Algorithms

EXPERIENCE

Backend AI Engineering Intern — *Flyrank AI* | 01-07-2026 – present

- **Engineered** backend services and RESTful APIs using Node.js and Express.js, supporting AI-agent workflows and MCP-based integrations for internal use cases.
- **Designed and deployed** an AI Study Coach using RAG, enabling students to upload study materials and ask context-aware questions based on their documents.
- **Built** a PDF report-generation pipeline integrated with SQLite to automate structured report generation
- **Collaborated** with senior engineers using Git/GitHub for version control, following Agile sprint cycles and code review practices.

PROJECTS

Adaptive Context-Aware DNS Resolver

Tech Stack: *Python, Flask, SQLite, React, REST APIs*

- **Engineered** an adaptive DNS resolver that dynamically selects the optimal DNS server based on real-time latency and availability metrics.
- **Designed and implemented** REST APIs for server health checks and lookup routing, tested via Postman.

Healthcare Appointment & Queue Management System

Tech Stack: *React.js, Flask, SQLite, Python*

- **Built** a full-stack healthcare platform supporting online appointment booking, AI-assisted patient prioritization, and real-time queue tracking.
- **Developed** REST APIs and a doctor-facing dashboard for appointment and queue management.

PDF Report Generator

Tech Stack: *Node.js, Express.js, SQLite, PDFKit*

- **Built** a backend service that queries student records from SQLite and auto-generates formatted PDF reports.
- **Automated** repetitive report preparation through database-driven PDF generation.

EDUCATION

B.E., Computer Science & Engineering (Artificial Intelligence & Machine Learning) Expected 2027

KLS Vishwanathrao Deshpande Institute of Technology | CGPA: 7.55 / 10

CERTIFICATIONS

- Anthropic AI Fluency
- IBM Machine Learning
- AWS Cloud Foundations
- Deloitte Data Analytics Job Simulation — Forage

ACHIEVEMENTS & LEADERSHIP

- 3rd Place — HACKFUSHION Hackathon (Built a gamified cybersecurity simulation in 20 hours with a 4-member team)
- Junior Operations Lead, Software Society — coordinated technical events/workshops
- Music Club Coordinator — organized and led events with 10+ team members